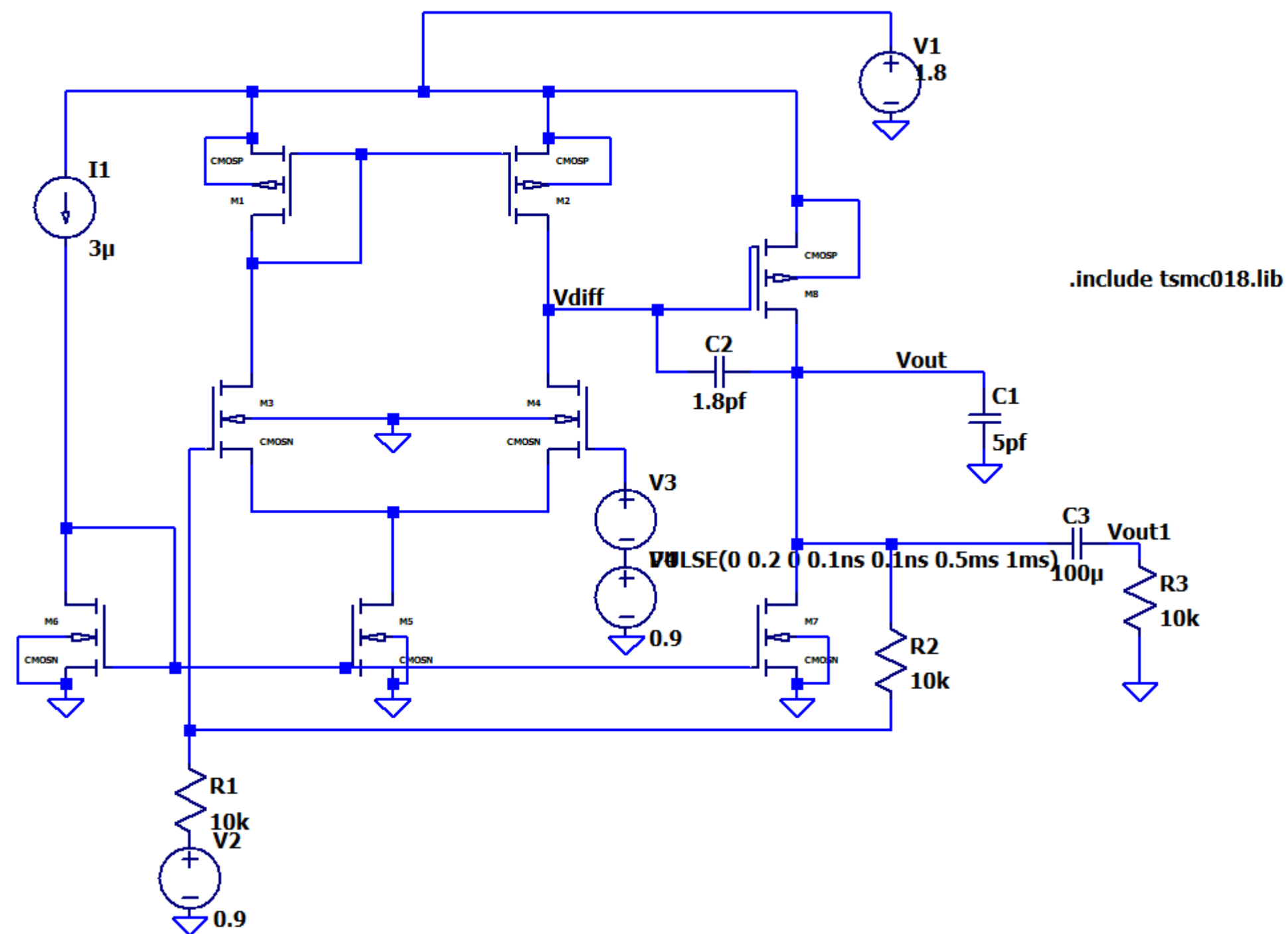
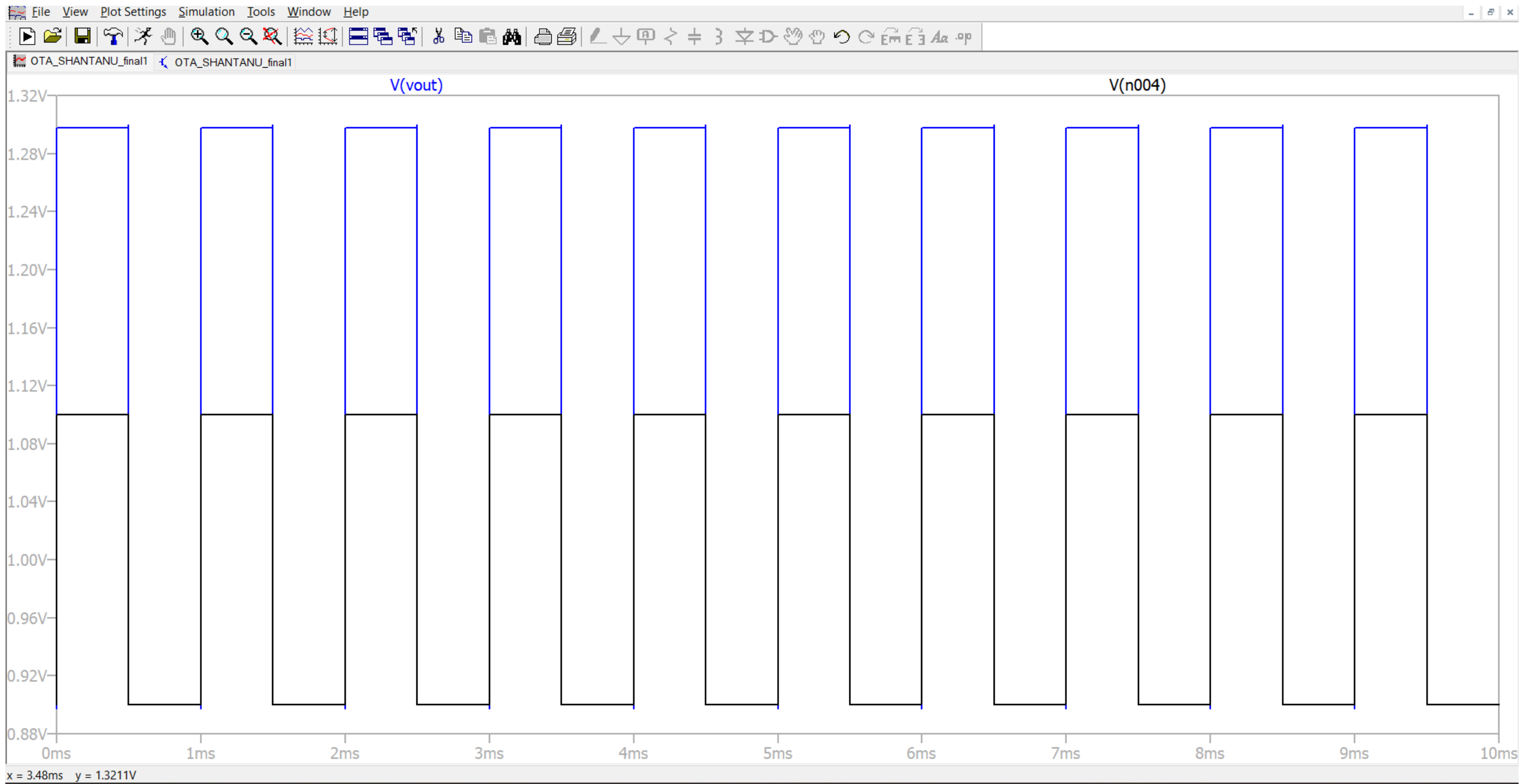


Non inverting
amplifier with
gain 2

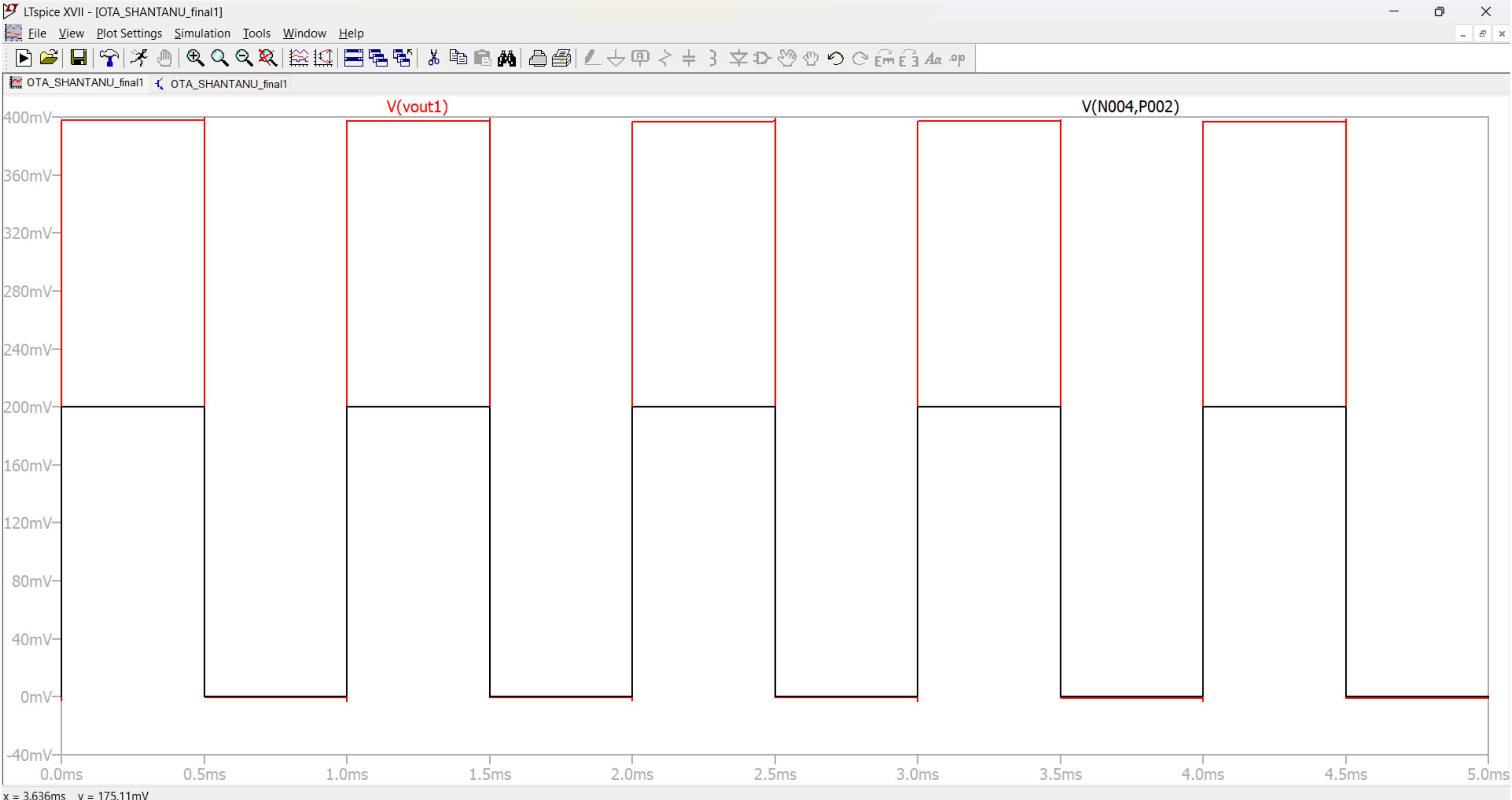


.tran 5ms

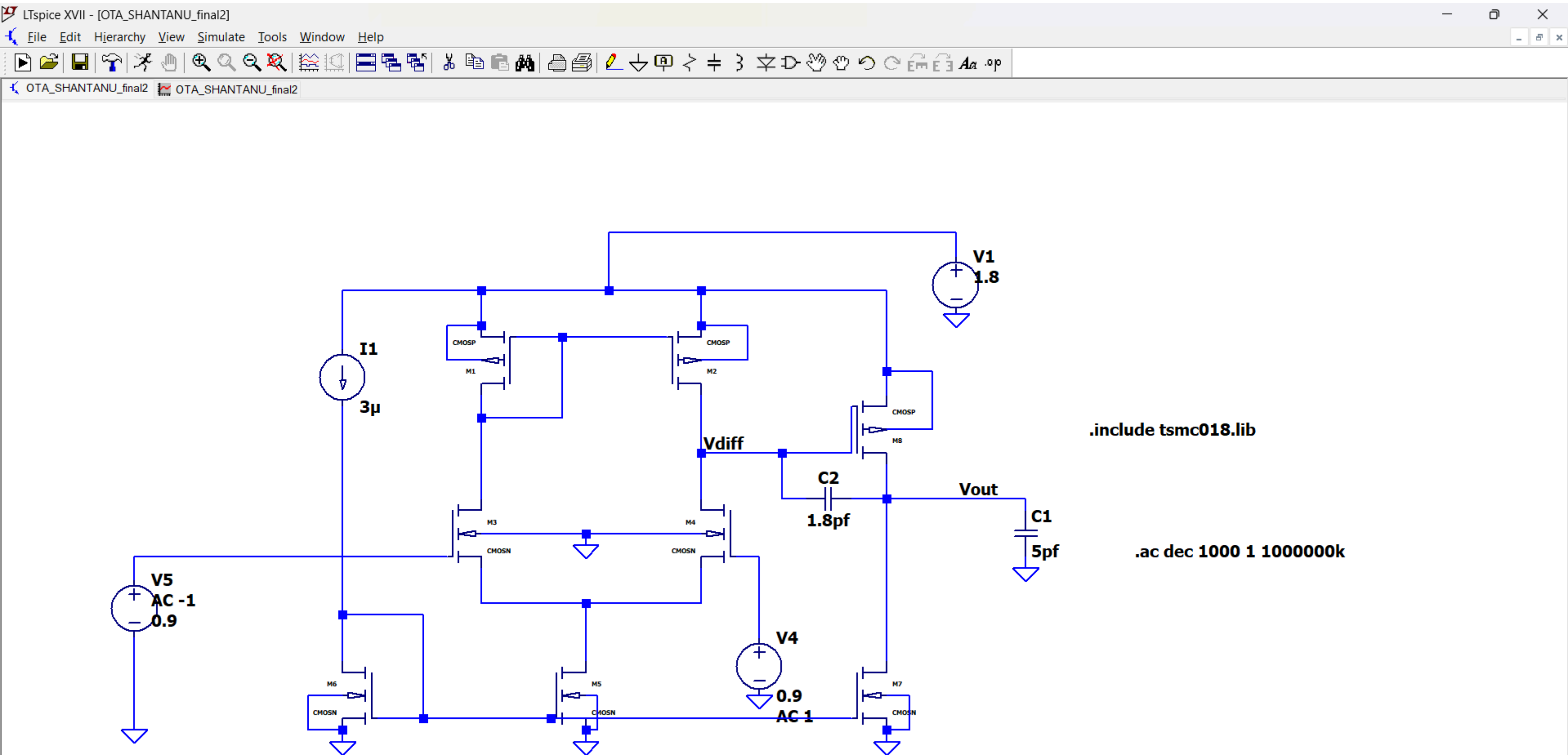
Simulation with Input pulse of 0.2V(black) giving output 0.4V(Blue).....at terminal Vout



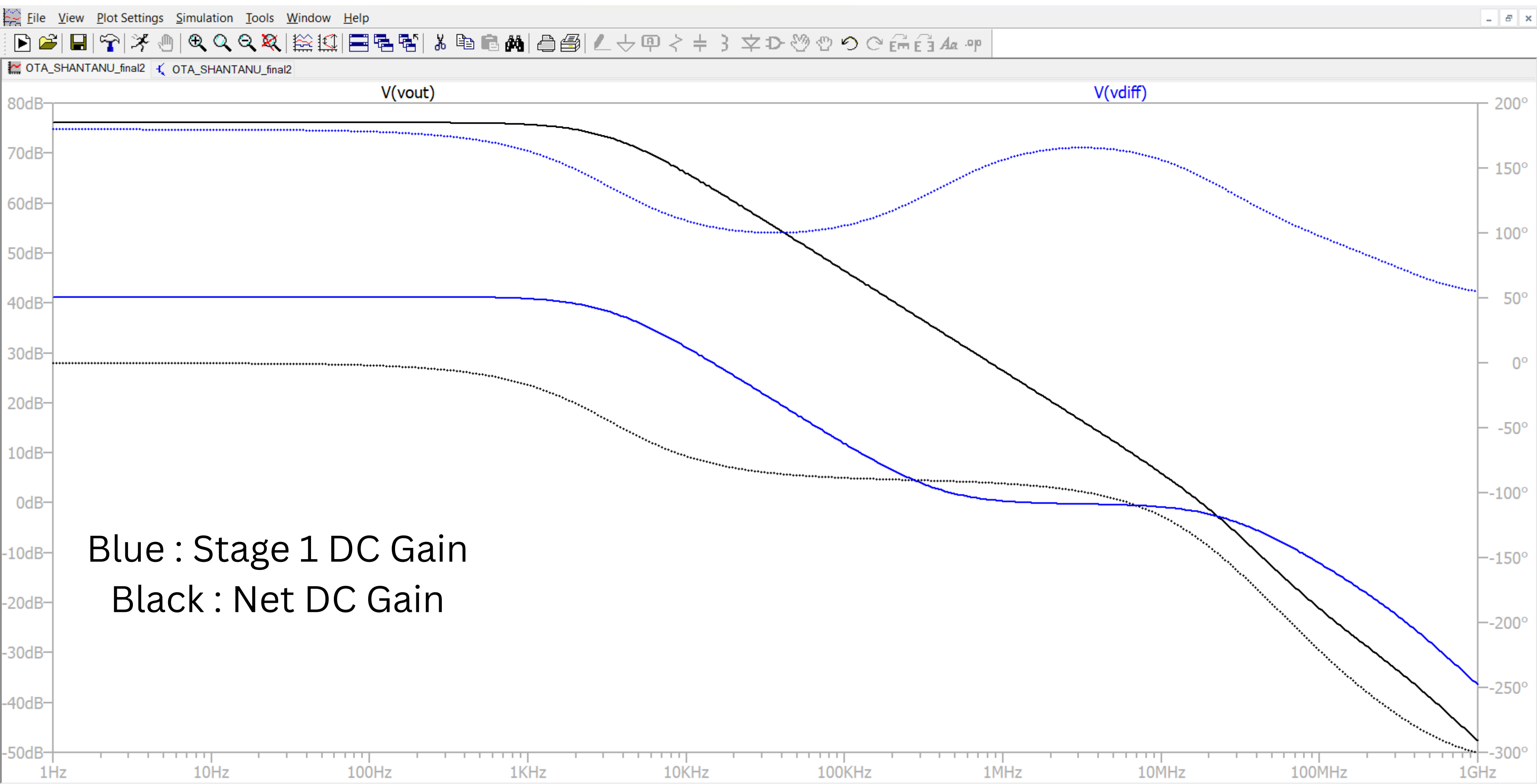
Output taken at terminal Vout1



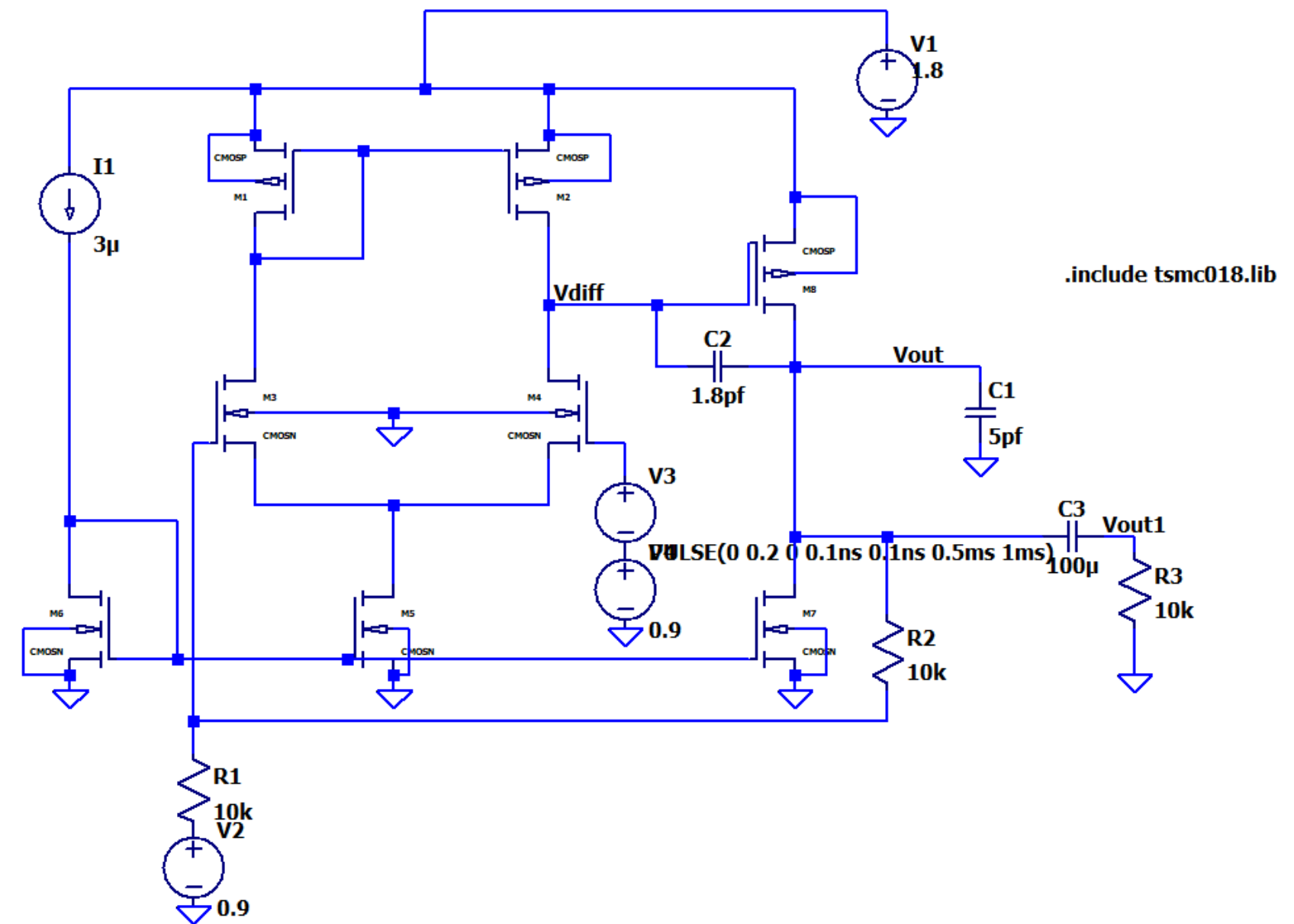
DC Gain



DC Gain

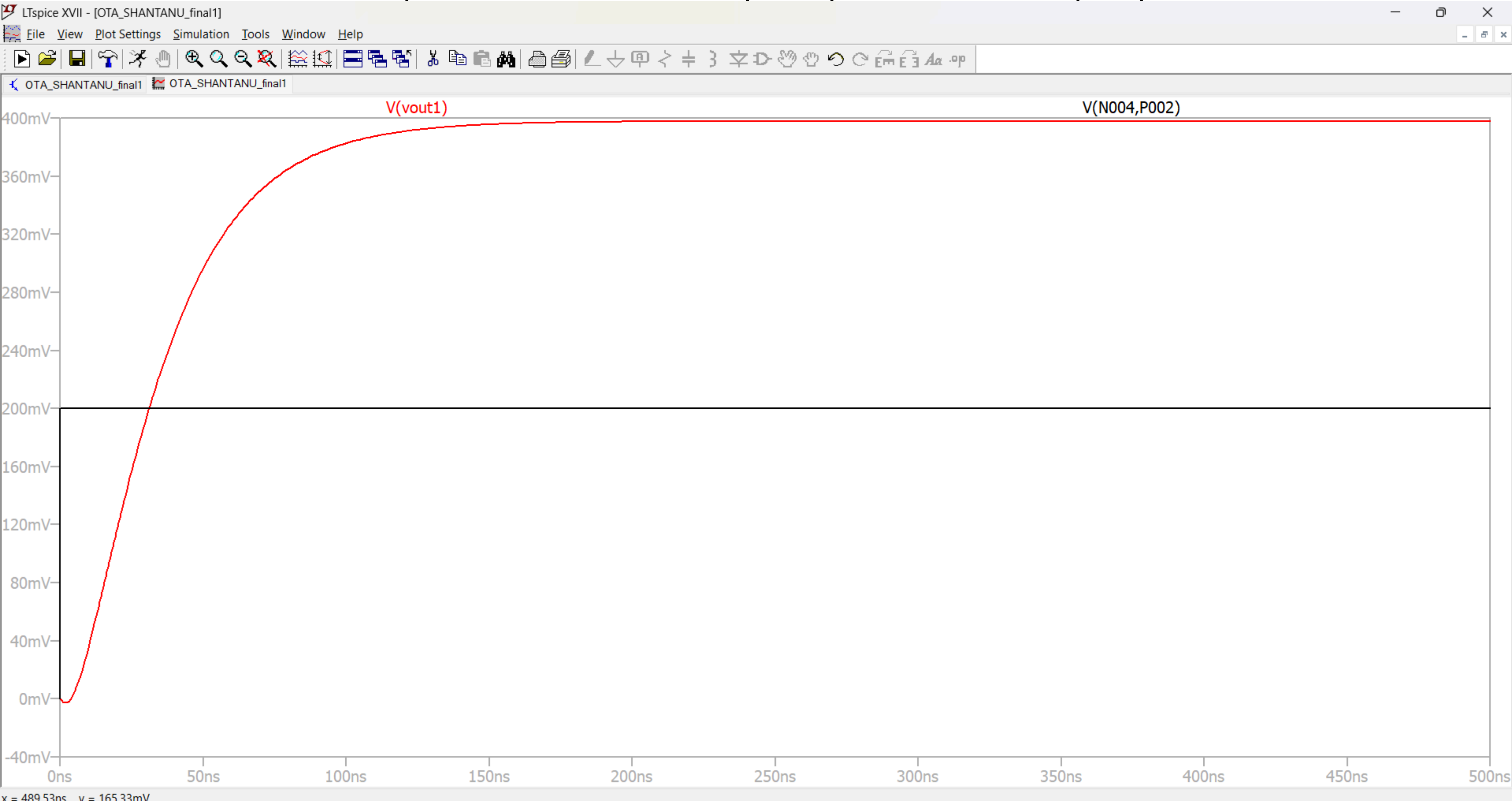


Transient Response of Closed loop amplifier with step input 0.2V

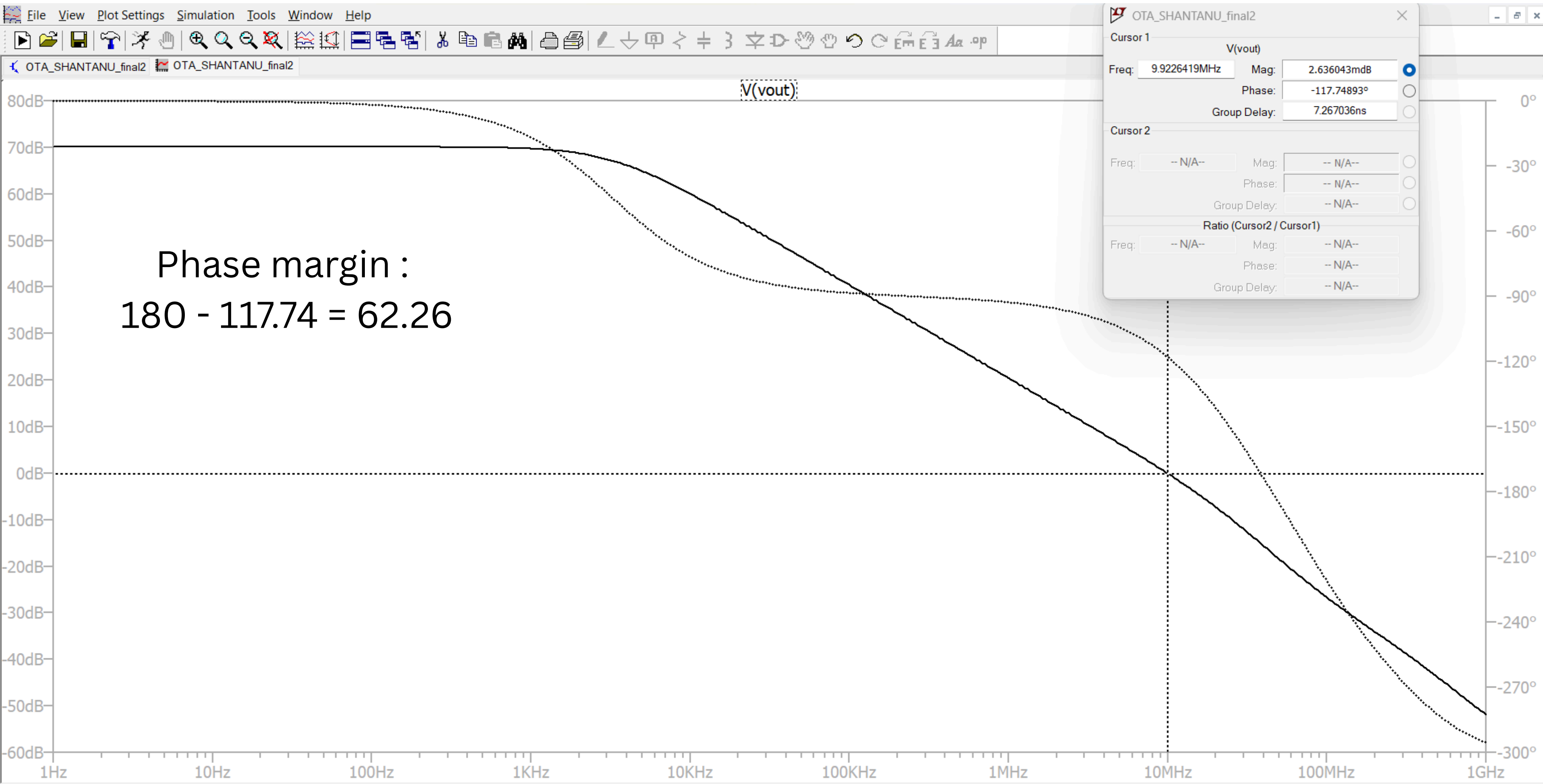


```
.tran 500ns
```

Transient Response of Closed loop amplifier with step input 0.2V



Phase Margin



DC operating points

 SPICE Error Log: C:\Users\shant\AppData\Local\Microsoft\Windows\INetCache\IE\NY7E9M5Z\OTA_SHANTANU_final2[1].log

Warning: Pd = 0 is less than W.

Warning: Ps = 0 is less than W.

Direct Newton iteration for .op point succeeded.

Semiconductor Device Operating Points:

--- BSIM3 MOSFETS ---

Name:	m8	m2	m1	m5	m7
Model:	cmosp	cmosp	cmosp	cmosn	cmosn
Id:	-9.68e-05	-1.02e-05	-1.02e-05	2.04e-05	9.68e-05
Vgs:	-6.21e-01	-6.21e-01	-6.21e-01	6.27e-01	6.27e-01
Vds:	-1.14e+00	-6.21e-01	-6.21e-01	2.45e-01	6.58e-01
Vbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Vth:	-4.40e-01	-4.41e-01	-4.41e-01	4.44e-01	4.43e-01
Vdsat:	-1.52e-01	-1.51e-01	-1.51e-01	1.41e-01	1.42e-01
Gm:	9.88e-04	1.05e-04	1.05e-04	2.10e-04	1.01e-03
Gds:	7.99e-06	1.14e-06	1.14e-06	8.45e-06	9.97e-06
Gmb	3.03e-04	3.19e-05	3.19e-05	5.84e-05	2.80e-04
Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgsov:	2.67e-14	2.93e-15	2.93e-15	2.02e-15	9.18e-15
Cgdov:	2.67e-14	2.93e-15	2.93e-15	2.02e-15	9.18e-15
Cgbov:	4.41e-19	4.41e-19	4.41e-19	4.66e-19	4.66e-19
dQgdVgb:	1.84e-13	2.02e-14	2.02e-14	1.22e-14	5.51e-14
dQgdVdb:	-2.64e-14	-2.90e-15	-2.90e-15	-2.04e-15	-9.04e-15
dQgdVsb:	-1.53e-13	-1.68e-14	-1.68e-14	-9.64e-15	-4.37e-14
dQddVgb:	-7.82e-14	-8.60e-15	-8.60e-15	-5.26e-15	-2.37e-14
dQddVdb:	2.65e-14	2.92e-15	2.92e-15	2.06e-15	9.11e-15
dQddVsb:	6.80e-14	7.47e-15	7.47e-15	4.22e-15	1.91e-14
dQbdVgb:	-2.71e-14	-2.99e-15	-2.99e-15	-1.65e-15	-7.73e-15
dQbdVdb:	1.87e-17	-1.91e-18	-1.91e-18	-5.78e-17	6.68e-18
dQbdVsb:	-9.90e-15	-1.07e-15	-1.07e-15	-8.13e-16	-3.67e-15

DC operating points

SPICE Error Log: C:\Users\shant\AppData\Local\Microsoft\Windows\INetCache\IE\NY7E9M5Z\OTA_SHANTANU_final2[1].log			
Name:	m6	m4	m3
Model:	cmosn	cmosn	cmosn
Id:	3.00e-06	1.02e-05	1.02e-05
Vgs:	6.27e-01	6.55e-01	6.55e-01
Vds:	6.27e-01	9.35e-01	9.35e-01
Vbs:	0.00e+00	-2.45e-01	-2.45e-01
Vth:	4.02e-01	5.11e-01	5.11e-01
Vdsat:	1.69e-01	1.24e-01	1.24e-01
Gm:	2.64e-05	1.27e-04	1.27e-04
Gds:	2.21e-07	1.06e-06	1.06e-06
Gmb:	7.36e-06	3.23e-05	3.23e-05
Cbd:	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00
Cgsbv:	4.03e-16	1.36e-15	1.36e-15
Cgdov:	4.03e-16	1.36e-15	1.36e-15
Cgbv:	9.66e-19	4.66e-19	4.66e-19
dQgdVgb:	4.13e-15	8.06e-15	8.06e-15
dQgdVdb:	-3.93e-16	-1.34e-15	-1.34e-15
dQgdVsb:	-3.55e-15	-6.40e-15	-6.40e-15
dQddVgb:	-1.73e-15	-3.48e-15	-3.48e-15
dQddVdb:	3.98e-16	1.35e-15	1.35e-15
dQddVsb:	1.76e-15	2.73e-15	2.73e-15
dQbdVgb:	-6.68e-16	-1.09e-15	-1.09e-15
dQbdVdb:	1.12e-19	1.66e-18	1.66e-18
dQbdVsb:	-3.71e-16	-4.17e-16	-4.17e-16
Date: Tue Apr 29 22:12:45 2025			
Total elapsed time: 0.046 seconds.			